# Institutional breadth as a cross-sectional signal: research memo

2026-11-08

This memo summarises the full technical report (`report/report.html`), which documents the data pipeline, the software, the design and every result in detail.

## 1. Question and signal

Does the change in the number of institutions holding a stock, read from Form 13F filings only after they are public, predict the cross-section of returns of liquid US stocks, net of costs, and does it add to momentum? The signal F1 is the change in breadth of ownership of Chen, Hong and Stein (2002, p. 180): among filers present in both quarters, holders at q minus holders at q-1, divided by the number of filers. It was chosen because it is defined in a peer-reviewed paper, needs only holdings counts (robust to value errors in 13F data), and connects to three mechanisms: gradual accumulation by institutions as information diffuses (an underreaction and momentum channel), informed institutional trading (Sias, Starks and Titman 2006), and short-sale constraints (Chen, Hong and Stein). The threat is that the signal was public for up to 45 days before formation, and that Nagel (2005) found the breadth effect reversed after 1998, before our sample starts. Four alternatives were considered and deferred (`docs/factor_memo_draft.md`, Section 3): persistent buying over several quarters, the breadth change of the best-performing filers, consensus best ideas, and crowding measured in days to liquidate. Each needs something F1 does not: a long-horizon return test, a skill ranking with its own timing problem, a benchmark weight per filer, or a reading of crowding as a risk premium rather than a signal.

## 2. Data and design

SEC Form 13F data sets (2013Q2 to 2026Q2) give holdings; Tiingo gives total-return daily prices for live and dead tickers; OpenFIGI maps CUSIPs to permanent identifiers. A `targets` pipeline moves data through bronze (as delivered), silver (point-in-time tables) and gold (signals and panels), with a data-quality suite before each pin.

- **Point in time.** A filing is usable the trading day after its filing date; amended reports are versioned so a reader at date t sees only the versions current at t, and a late report enters at the first formation after it is public. A random sample of 300 filings checked against EDGAR's acceptance times confirms that every one was public before the close of its usable date, when the portfolio trades.
- **Amendments and confidential treatment.** A restatement replaces its report and a NEW HOLDINGS amendment adds positions to it, each from its own usable date. Positions withheld under a confidential treatment request (declared on about 113 original reports a quarter) are invisible until an amendment reveals them, and that amendment enters only from its own usable date, so nothing withheld is used early. The cost is that a withheld position reads as an exit until it is revealed.
- **Portfolios.** F1 quintiles, formed at the end of February, May, August and November and held for three months with weights drifting in between; returns are measured monthly. Three weighting schemes: cs (signal score times a liquidity score, pre-registered), ew (equal weights) and rp (risk parity). The spread is the top minus the bottom quintile, one dollar long and one dollar short.
- **Evaluation.** The alpha of the monthly spread against the Fama-French five factors plus momentum (UMD), called FF6 alpha, with Newey-West standard errors; the rank information coefficient (IC) between F1 and next-quarter returns; a double sort within momentum terciles; a monotonicity test; a terminal-loss stress; and fees of 10 to 50 bps per dollar traded plus a 100 bps borrow fee. The long legs are also compared with SPY and with an equal-weight portfolio of the whole universe. A credible signal must pass under all three weighting schemes.

**Scope and what was excluded.** Filers: every 13F filer, then at each formation only those with a report for both quarters compared and at least 10 positions in each. Index-like filers (top third by number of positions and bottom third by turnover) are removed, because their holdings change with index membership and fund flows rather than views: about 509 filers per formation, 9% of filers but 52% of 13F dollar value. When one report includes another manager that also files alone, the smaller report is dropped. Securities: US common stocks only (ETFs, ADRs, funds and options are out of scope); at each formation the 1,000 most liquid with a price above USD 5, a year of history, trading on 90% of recent days and a mapping confirmed by the holders' own prices. Window: report quarters 2013Q2 to 2026Q2. Short positions are not reported on 13F and are not modelled.

| Filers at a formation | Average number | Share of filers | Share of 13F value |
|---|---|---|---|
| Cohort member | 4,162 | 76.3% | 44.2% |
| Fewer than 10 positions | 618 | 11.4% | 0.8% |
| Index-like | 509 | 9.4% | 51.8% |
| Included in another filer's report | 166 | 3.0% | 3.1% |

Candidate stocks are all those held by the cohort, most of them small: on average 18% of candidate stock-formations enter the universe, 31% pass every screen but rank below the 1,000 most liquid, and 27% fail the identity or price-source checks.

### 2.1. Point-in-time discipline

The rule does not depend on the 45-day deadline. (1) Each filing gets a usable date, the first trading day after its filing date. (2) A report and its amendments form versions, each valid from the usable date of the filing that created it until the one that replaces it, so a reader at date t sees exactly the versions with valid_from <= t < valid_to. (3) At formation t the cohort, the holder counts and the overlap links use only those versions, for both quarters compared, and the portfolio trades at the close of t. (4) A late report enters at the first formation after its own usable date; until then its filer is outside the cohort, and a missing report is never read as an exit. Every price feature uses trading days before t, the USD 5 floor uses the raw close, and returns start the day after t.

Filers file close to the deadline: 25.7% of original reports are usable within 30 days of quarter end, 66.3% within 45 and 96.7% within 60. Formations at the end of February, May, August and November fall about two weeks after the deadlines, so most reports of the quarter are in; the rest wait for the next formation.

| Audit of 300 random filings against EDGAR | Result |
|---|---|
| EDGAR index pages read | 300 of 300 |
| Bulk filing date equals EDGAR filing date | 100.0% |
| Accepted before the usable date | 99.7% |
| Public before the close of the usable date | 100.0% |

The one filing accepted after its filing date is an amendment that EDGAR dated 2020-09-01 and accepted at 9:01 am on 2020-09-02, its usable date, seven hours before the close; the pipeline stops if any sampled filing was accepted after that close.

## 3. Results

| Spread | Mean, % a year | CAPM alpha | t | FF6 alpha | t | 95% CI of FF6 alpha |
|---|---|---|---|---|---|---|
| F1, cs (pre-registered) | 4.48 | 6.22 | 1.84 | 2.25 | 1.06 | [−1.93, 6.42] |
| F1, equal weight | 3.10 | 5.24 | 2.05 | 1.61 | 0.94 | [−1.74, 4.95] |
| F1, risk parity | 2.46 | 3.41 | 1.70 | 0.61 | 0.41 | [−2.28, 3.50] |
| F1 within momentum terciles | 3.39 | 3.54 | 1.61 | 2.38 | 1.44 | [−0.86, 5.63] |
| F1, terminal stress | 4.88 | 6.59 | 1.95 | 2.67 | 1.26 | [−1.48, 6.83] |
| F1 net, 25 bps + 100 bps borrow | 0.71 | 2.26 | 0.67 | −1.81 | −0.84 | [−6.02, 2.40] |
| Momentum (reference) | 5.22 | 9.05 | 1.65 | 1.19 | 0.39 | [−4.72, 7.10] |

1. **Predictive content: no evidence.** The cs spread earns 4.48% a year gross, with an FF6 alpha of 2.25% and an interval of [−1.93, 6.42]. The mean rank IC is 0.011 (t = 0.74). Returns are flat from the first to the fourth quintile and rise only at the fifth; the monotonicity test does not reject a flat relation (p = 0.53).
2. **Economic viability: viable on the point estimate under cs only, with a wide interval.** Net of a 25 bps fee and 100 bps borrow, the cs spread earns 0.71% a year, 95% interval [−6.83, 8.25]. The pre-registered rule calls a positive

net return viable, with its interval reported; the interval includes zero, and under ew and rp the same spread earns −0.87% and −1.60%. The break-even fee is 40 bps per dollar traded.

3. **Incremental to momentum: positive in direction, not significant.** Within each momentum tercile, F1 still earns a positive spread, 3.39% a year on average with the highest Sharpe ratio of any test (0.37), but its alpha interval [−0.86, 5.63] includes zero, so the pre-registered verdict is "not shown".

The weighting hurdle is not met: the spread is positive under all three schemes but no FF6 alpha is significant. Against the market alone, the ew spread has a CAPM alpha with t = 2.05; the other five factors absorb most of it.

All alphas are estimated on 151 months, because the Fama-French data end in September 2026, while the means use all 152 months. The month left out, October 2026, was the worst of the sample: the cs F1 spread lost 20.0% and the momentum spread 18.3%. On the common months the alphas are therefore higher than they would be with October included. The CAPM alpha removes only the market; the legs have almost equal market betas, so it differs from the mean mainly through that missing month, and the gap between the CAPM and FF6 alphas shows what the other five factors explain.

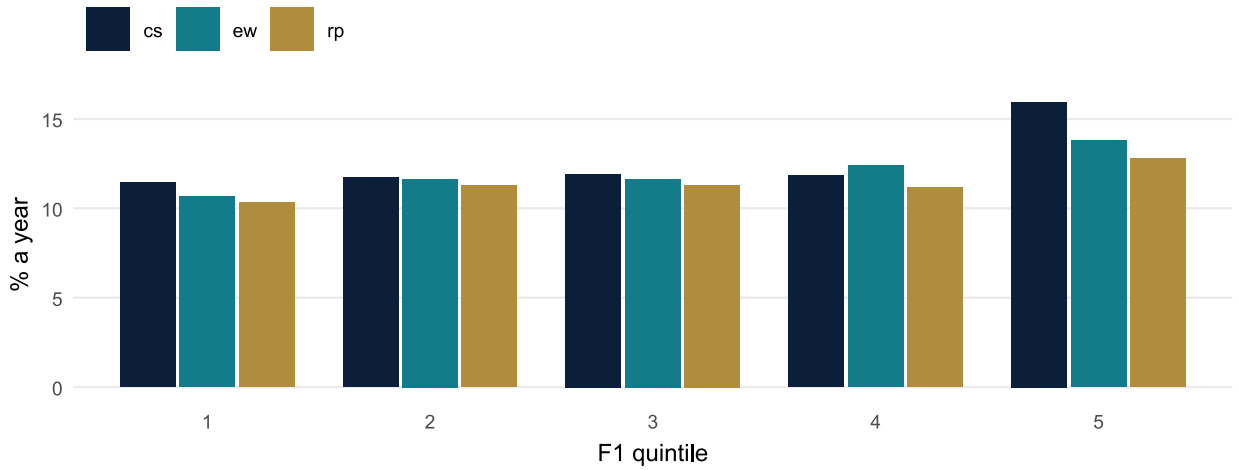



Figure 1: Annual return of each F1 quintile (1 = lowest breadth change).

The extreme quintiles hold more liquid stocks than the middle ones even with equal weights (liquidity percentile 0.55 and 0.64 against about 0.44): the extremes of F1 are widely held stocks, which can gain or lose hundreds of holders in a quarter. At the last formation the three largest F1 values were INTC, MU, AMD, adding 708 to 863 holders each.

## 3.1. Costs

| Weights | Gross, % a year | Net at 10 bps | Net at 25 bps | Net at 50 bps |
|---|---|---|---|---|
| cs | 4.48 | 3.37 | 1.71 | −1.06 |
| ew | 3.10 | 1.91 | 0.13 | −2.85 |
| rp | 2.46 | 1.23 | −0.60 | −3.65 |

The cost model charges a fee on every dollar traded at a rebalance (10, 25 or 50 bps, pre-registered) and a borrow fee of 100 bps a year on the short leg. factoRverse can also charge market impact, which grows with an order's size relative to the stock's daily dollar volume; it is switched off here because its size depends on the assets managed, which this study does not fix. Costs absorb most of the gross spread. Under cs weights the bottom leg trades 1.47 times its value at each rebalance and the top leg 1.30 times (purchases plus sales). F1 has little persistence: its rank correlation between consecutive formations averages 0.11. A borrow fee of 100 bps a year on the short leg costs a further 1 percentage point. A signal averaged over two quarters may keep more of each quintile and trade less; that is a hypothesis for the next step, not a result.

## 3.2. F1 and momentum

| Gross spread, cs | Mean, % a year | Volatility, % | Sharpe | FF6 alpha | t |
|---|---|---|---|---|---|
| F1 among past losers | 1.84 | 10.97 | 0.17 | 1.47 | 0.64 |

| Gross spread, cs | Mean, % a year | Volatility, % | Sharpe | FF6 alpha | t |
|---|---|---|---|---|---|
| F1 in the middle tercile | 2.41 | 9.52 | 0.25 | 2.11 | 0.91 |
| F1 among past winners | 5.93 | 14.83 | 0.40 | 3.57 | 1.25 |
| F1 within terciles, average | 3.39 | 9.13 | 0.37 | 2.38 | 1.44 |
| Momentum spread | 5.22 | 24.73 | 0.21 | 1.19 | 0.39 |

The double sort holds momentum roughly fixed and asks whether F1 still ranks returns. It does, in direction: the F1 spread is positive in every momentum tercile and grows with momentum, from 1.84% a year among past losers to 5.93% among past winners. Averaged over the three terciles it has a Sharpe ratio of 0.37, above both the unconditional F1 spread (0.29) and the momentum spread (0.21), and the alpha closest to significance of all the tests (t = 1.44). Rising breadth among stocks that have already risen is what the gradual-trend reading of F1 predicts. None of these spreads is significant, and the winners tercile is one of three, noticed after seeing results, so it is a hypothesis for the next trial rather than a finding.

The unconditional F1 and momentum spreads move together (correlation 0.86), yet adding F1 helps a momentum investor: half in each gives a Sharpe ratio of 0.25 against 0.21 for momentum alone, because F1 earns most of momentum's return with less volatility (15.7% against 24.7%). These comparisons are descriptive and gross of costs.

### 3.3. Benchmarks

| Long-only portfolio, gross | Mean, % a year | 95% interval |
|---|---|---|
| SPY | 13.85 | [7.45, 20.25] |
| Equal-weight universe | 12.23 | [3.88, 20.58] |
| Top F1 quintile, cs | 15.94 | [6.59, 25.30] |
| Bottom F1 quintile, cs | 11.46 | [2.16, 20.76] |

The spread is a zero-investment portfolio, judged against zero and the factors. The long legs are compared with SPY, the default holding of a US equity investor, and with an equal-weight portfolio of the same 1,000 stocks: the top quintile earned more than both before costs, but the intervals overlap widely.

## 4. How to read the result

The test has little power. With 152 months of returns (151 in the factor regressions, as the factor data stop one month earlier), a true FF6 alpha would have to be about 6.0% a year to be detected four times out of five, and a true mean spread about 10.8%; the pre-registered threshold of economic interest was a 2% alpha. Momentum, the closest known anomaly, was itself insignificant in the window: UMD averaged 2.66% a year (t = 0.76) against 7.25% since 1963. The evidence is therefore inconclusive, not a rejection, and it is consistent with Nagel's finding that the breadth effect weakened after 1998.

| UMD | Mean, % a year | t | CAPM alpha | t |
|---|---|---|---|---|
| study window | 2.66 | 0.76 | 6.00 | 2.05 |
| since 1963 | 7.25 | 3.98 | 8.36 | 4.96 |

## 5. Design choices and deviations

The hypotheses, definitions and decision rules were fixed before any return was computed, in `docs/factor_memo_draft.md` (hypothesis in its Section 3, decision rules in Section 8); every later change is dated in that document's Deviations section. The main ones:

- **Identity at t.** A CUSIP is accepted only if most holders report a value per share close to Tiingo's close at quarter end, recomputed at each formation.
- **Suspected CUSIP changes.** A stock where more than half of at least 20 holders are new is excluded for that formation: corporate actions that move holders to a new CUSIP otherwise look like mass initiations.
- **Denominator.** F1 divides by the matched cohort rather than all funds; ranks within a formation are identical.

- **Traded-day presence.** The universe requires trading on 90% of the previous 63 days, so that a stock whose vendor price stays frozen after a takeover, delisting or halt cannot enter it.
- **Short-leg weighting.** Under cs the bottom quintile is weighted by conviction in the short direction (negated F1 times liquidity).

## 6. Reproducibility

The project is an R package with a `targets` pipeline: `renv::restore()` then `targets::tar_make()` rebuilds every pin, the report and this memo. Each layer pins only after its data-quality suite passes; over 300 unit tests run before every commit, and continuous integration checks the package on a clean machine. The backtest engine, factoRverse, ships in the repository.

## 7. Limitations

Short sample and low power; a structural size tilt in F1; identity from today's OpenFIGI and Tiingo mappings (of 13F dollar value, 71.7% is priced, 24.2% is out of scope or unmapped, 3.7% is common stock with no Tiingo price series, and 0.45% is removed by the identity checks, mostly dead firms whose ticker was later reused); formation up to two weeks after most filings are public; stocks that stop trading after formation earn zero until the next formation. The size tilt matters most: F1 partly ranks stocks by how widely they are held.

## 8. Next steps

In priority order:

1. **F1b and F1c.** The ownership change by value and the number of new holders relative to the stock's own holders measure the same idea without the size tilt of F1, the main weakness of the current test. Each is a separate trial, with false discovery rate control across the three signals and a Sharpe ratio deflated for the number of trials.
2. **F1 among past winners.** The strongest pattern in the results is the F1 spread within the high-momentum tercile. Tested as a pre-registered conditional strategy, counted as a new trial, with leave-one-year-out and sector checks, it would show whether rising breadth confirms a price trend or the pattern is noise.
3. **Costs.** Turnover decides whether any version is tradable. A smoothed F1 (two quarters), factoRverse's market impact model by strategy size, and realistic borrow fees would show the cost a real portfolio would pay.